Conjecture 00000001491: the printed map collapses onto a line

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1 The explicit formula matters

Both source languages explicitly define

$$H_{a,b}(x, y) = (1 - ax^2 + y,\ b).$$

The second coordinate here is the constant b, not bx. The usual Hénon map in Hénon's original abstract [1] has second coordinate bx. Our result concerns exactly the formula printed in this problem; it does not assert a dimension theorem for that different, usual Hénon map.

For the printed map, every image lies on the horizontal line $L_b = \{(x, y) : y = b\}$. Consequently every positive-time iterate, every actual forward ω-limit set, and the closure of the union of all actual point ω-limit sets lie on L_b. Their Hausdorff dimensions are at most one. Any compact fully invariant set $H_{a,b}(A) = A$ also has dimension at most one. These statements are proved for all real a, b, then instantiated at the exact printed parameters $a = 7/5 = 1.4$, $b = 3/10 = 0.3$.

Thus the required lower bound $1.26 < d_H$ cannot hold for the printed map's actual limiting sets or a fully invariant compact attractor. This necessary lower-bound clause already refutes the conjunction. We do not calculate the usual Hénon attractor's dimension or assert independent results about box dimension or the Kaplan–Yorke formula. No classification of all alternative Milnor-style attractor definitions is claimed.

2 The actual Euclidean plane and horizontal line

Lean uses the actual Euclidean space $E = \mathrm{EuclideanSpace}(\mathbb{R}, \mathrm{Fin}\, 2)$, with its Euclidean metric, and the actual continuous polynomial map given above. The line $L_b \subset E$ is closed because the second coordinate projection is continuous. For every $v \in E$,

$$H_{a,b}(v) \in L_b, \qquad H_{a,b}^n(v) \in L_b \quad (n \geq 1).$$

The second statement follows from the actual iterate successor identity, not an assumed finite-state or line-restricted model.

The genuine line parametrization

$$\iota_b : \mathbb{R} \longrightarrow E, \qquad t \longmapsto (t, b)$$

is an actual isometry onto L_b. Lean proves the distance identity using the Euclidean squared-coordinate formula, proves $L_b = \iota_b(\mathbb{R})$, and applies isometric invariance of Hausdorff dimension and $\dim_H \mathbb{R} = 1$. Therefore

$$\dim_H L_b = 1, \qquad A \subseteq L_b \Rightarrow \dim_H A \leq 1.$$

No max-norm substitution or unproved metric conversion is used.

3 Actual forward limit sets

For any actual initial set $S \subset E$, define

$$\Omega_{a,b}(S) = \mathrm{omegaLimit}\big(\mathrm{atTop},\ (n, v) \mapsto H_{a,b}^{n}(v),\ S\big).$$

This is Mathlib's actual filter-based ω-limit set: the intersection, over $U \in \mathrm{atTop}$, of the closures of $\{H_{a,b}^{n}(v) : n \in U, v \in S\}$. Take the actual filter member $U = \{n : n \geq 1\}$. All these images lie in the closed line L_b, so their closure does too. Hence

$$\Omega_{a,b}(S) \subseteq L_b, \qquad \dim_H \Omega_{a,b}(S) \leq 1 \quad \text{for every } S \subset E.$$

In particular this applies to every single orbit $S = \{v\}$. The separately defined actual global limiting object

$$K_{a,b} = \overline{\bigcup_{v \in E} \Omega_{a,b}(\{v\})}$$

also lies in L_b by closedness, and has Hausdorff dimension at most one. No arbitrary set is simply named an attractor in place of these actual dynamical objects.

4 Full invariance and the real dimension bound

For every actual set $A \subset E$ with $H_{a,b}(A) = A$, each $v \in A$ is an actual image $H_{a,b}(w)$ for some $w \in A$. Thus $A \subseteq L_b$. In particular there is no compact fully invariant set with dimension greater than $63/50 = 1.26$. This uses full invariance; merely forward invariance $H_{a,b}(A) \subseteq A$ would not suffice, since a thick trapping region may contain points off the line. We do not use that weaker condition as a substitute for an attractor.

Mathlib's `dimH` takes values in the extended nonnegative reals. Lean first proves the actual bound $\dim_H A \leq 1$ for every line subset, and separately proves $\dim_H A < \infty$. The real-value bridge then gives $\mathrm{toReal}(\dim_H A) \leq 1$. The final theorem

`source_parameters_real_dimensions`

explicitly gives this real bound for every actual $\Omega_{7/5,3/10}(S)$ and for $K_{7/5,3/10}$, and denies existence of a compact fully invariant A with real dimension above $63/50$. There is no use of an infinite-dimension conversion default.

5 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits of all 21 theorems and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.

References

[1] M. Hénon, *A two-dimensional mapping with a strange attractor*, Commun. Math. Phys. **50** (1976), 69–77. `https://doi.org/10.1007/BF01608556`.